Measuring performance. For binary classification, the confusion matrix counts true positives, false positives, true negatives, and false negatives. Accuracy is the fraction of correct predictions, but it can be misleading on imbalanced data. Precision is the fraction of predicted positives that are truly positive, and recall is the fraction of actual positives that are found. The F1 score is the harmonic mean of precision and recall. The ROC curve plots the true positive rate against the false positive rate as the decision threshold changes, and the area under it, ROC AUC, summarizes ranking quality across all thresholds. For regression, common measures are the mean squared error and the mean absolute error.

Regularization and training techniques for deep networks. Dropout randomly sets a fraction of activations to zero during training, which discourages units from relying on each other and acts as an approximate ensemble. Batch normalization normalizes each layer's inputs over the mini-batch and then rescales them with learned parameters, which stabilizes and speeds up training. Early stopping halts training when validation loss stops improving. Data augmentation creates new training examples by transforming existing ones, such as cropping or flipping images. Transfer learning starts from a model pretrained on a large dataset and fine-tunes it on a smaller target dataset, which usually needs far fewer labelled examples than training from scratch.